Evaluating South Africa’s Regulatory Readiness for Software as a Medical Device (SaMD) and AI as a Medical Device (AIaMD): A Framework-Based Gap Analysis, Stakeholder Assessment, and Operational Recommendations

1. Title

Evaluating South Africa’s Regulatory Readiness for Integrating SaMD and AIaMD: Framework-Based Gap Analysis, Stakeholder Assessment, and Operational Recommendations

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2. Background and Motivation

Software-driven medical technologies—ranging from standalone diagnostic applications to machine learning algorithms for image analysis—are transforming healthcare globally (Ravishankar et al., 2020). Collectively known as **Software as a Medical Device (SaMD)** and **AI as a Medical Device (AIaMD)**, these innovations often fall under regulatory frameworks that were originally designed for physical, hardware-based devices (Medicines and Related Substances Act 101 of 1965 (as amended), 1965; SAHPRA, 2021).

In leading jurisdictions, regulators are adapting policies to address unique software and AI risks. For example, the U.S. Food and Drug Administration (FDA) has proposed **Good Machine Learning Practice (GMLP)** principles, which emphasize multidisciplinary design, robust software engineering, representative training data to avoid bias, and continuous monitoring of deployed algorithms (FDA, 2021). Similarly, the EU’s **Medical Device Regulation (EU MDR 2017/745)** frames rigorous standards for lifecycle management, clinical validation, and traceability, with emerging guidance specific to AI-driven tools (European Commission, 2017). The World Health Organization (WHO) highlights essential regulatory considerations such as algorithmic transparency, data privacy, fairness, and post-market surveillance within its Global Model Regulatory Framework for Medical Devices (WHO, 2017).

In **South Africa**, the current legal foundations for medical device oversight are set out in the **Medicines and Related Substances Act 101 of 1965** (as amended). The South African Health Products Regulatory Authority (SAHPRA) issues guidelines for device classification and licensing (SAHPRA, 2021; SAHPRA, 2023). However, these guidelines do not yet address the continuous learning nature of certain AI applications, nor do they set clear policies on bias mitigation, explainability, or cybersecurity for SaMD (Maidment, 2023). Consequently, innovators face uncertainty about **regulatory compliance**, while stakeholders (research ethics committees, healthcare professionals, manufacturers) lack clarity on best practices.

Against this backdrop, **this research** aims to systematically evaluate South Africa's regulatory preparedness for SaMD and AIaMD and propose a context-sensitive roadmap for modernization. The analysis will draw on international best practices—such as **FDA's GMLP**, **EU MDR**, **ISO 13485**, and **IEC 62304**—to identify key governance criteria and then assess their presence (or absence) in South Africa's current framework.

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3. Problem Statement

South Africa's regulatory approach to medical devices was historically geared toward hardware-based technologies, leaving **software-specific and AI-specific risks** insufficiently addressed (Medicines and Related Substances Act, 1965; SAHPRA, 2021). Issues like **algorithmic bias, drift, explainability, data security, and continuous learning** are not clearly regulated, despite their criticality to patient safety and public health (Challen et al., 2019; Bayram and Alatas, 2021). As a result, there is an evident **regulatory gap** regarding SaMD and AIaMD oversight. This gap not only poses risks for patient safety but also stifles local innovation by failing to offer a transparent path to market.

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4. Research Aim

The aim of this study is **to critically assess South Africa's regulatory readiness for SaMD and AIaMD and to develop an evidence-based compliance framework** that addresses identified gaps. By benchmarking against global best practices (FDA, 2021; European Commission, 2017; WHO, 2017; ISO 13485:2016; IEC 62304:2006) and capturing stakeholder insights, the project seeks to articulate a practical set of recommendations that can guide policy-makers, regulators, and industry stakeholders in **enhancing regulatory clarity and ensuring patient safety**.

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5. Research Objectives

1. **Framework-Based Gap Analysis**

    - Conduct a literature review of **international best practices** (e.g., FDA GMLP, EU MDR, WHO guidelines, ISO 13485, IEC 62304) and derive specific governance criteria for SaMD/AIaMD (FDA, 2021; European Commission, 2017; WHO, 2017; ISO 13485:2016; IEC 62304:2006).

    - Evaluate the **Medicines and Related Substances Act** and **SAHPRA guidelines** against these criteria to identify explicit shortcomings (SAHPRA, 2021; SAHPRA, 2023).

2. **Stakeholder Consultation**

    - Develop a **structured questionnaire and interview protocol** for stakeholders, including IRBs, SAHPRA, the South African Medical Research Council (SAMRC), contract research organizations (CROs), and university researchers (Maidment, 2023).

    - Capture stakeholder perceptions, experiences, and expectations of current regulatory oversight for SaMD/AIaMD in South Africa.

3. **Comparative Analysis**

    - Compare **documentary gap analysis** with **stakeholder feedback** to highlight areas of convergence and divergence.

    - Determine the **most critical gaps** and the **practical feasibility** of bridging them.

4. **Recommendations and Operational Tool**

    - Formulate **policy recommendations** and propose an **operational compliance tool** (e.g., a checklist) aligned with **global frameworks** and **local context** (WHO, 2017; FDA, 2021).

    - Ensure the recommendations address **bias mitigation, algorithmic transparency, cybersecurity, continuous learning updates, and post-market monitoring** for AI-driven devices (Challen et al., 2019; Bayram and Alatas, 2021).

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6. Research Questions

- **RQ1**: What specific **regulatory gaps** exist within South Africa's current medical device framework regarding accountability, fairness, transparency, human oversight, and technical robustness for SaMD/AIaMD?

- **RQ2**: How do **stakeholders** (IRBs, SAHPRA, SAMRC, CROs, academia) perceive existing regulations, and where do their experiences converge or diverge from global best practices (ISO 13485:2016; IEC 62304:2006)?

- **RQ3**: What **practical strategies** can address these gaps to enhance South Africa's readiness for SaMD/AIaMD, and how can an **operational compliance tool** facilitate alignment with international standards (FDA, 2021; European Commission, 2017)?

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7. Methodology

7.1 Literature Review and Criteria Development

A **comprehensive literature review** will first identify key regulatory elements relevant to SaMD/AIaMD. Sources include:

- **Regulatory frameworks**: FDA's GMLP (FDA, 2021), EU MDR (European Commission, 2017), WHO guidelines (WHO, 2017).
- **Standards**: ISO 13485:2016 (on quality management systems), IEC 62304:2006 (on software life cycle processes).
- **Scholarly research**: On algorithmic bias (Challen et al., 2019), drift detection (Bayram and Alatas, 2021), and AI-based medical imaging (Ravishankar et al., 2020).

From these, a set of **governance criteria** will be derived. Likely dimensions include **accountability and liability**, **fairness and bias mitigation**, **technical robustness**, **algorithmic transparency**, **cybersecurity**, **post-market surveillance**, and **human oversight** (FDA, 2021; WHO, 2017).

7.2 South African Regulatory Gap Analysis

Next, the researcher will analyze:

- **Medicines and Related Substances Act 101 of 1965 (as amended)**.
- **SAHPRA guidelines** on device classification, licensing, and safety (SAHPRA, 2021; SAHPRA, 2023).

Using the governance criteria as a benchmark, each document will be evaluated to identify **explicit references to AI/ML**, requirements for **continuous learning updates**, **cybersecurity**

provisions, and more. This **documentary analysis** will reveal where current texts fall short in meeting international best practices (European Commission, 2017; FDA, 2021).

7.3 Stakeholder Engagement

A structured **questionnaire** and optional **follow-up interviews** will be administered to:

- **Institutional Review Boards (IRBs)** and **Research Ethics Committees** (evaluating clinical research for new devices).
- **SAHPRA** officials (responsible for regulating medical devices).
- **SAMRC** representatives (involved in national health research).
- **Contract Research Organizations (CROs)** and **university researchers** developing or testing AIaMD.

The questionnaire will collect **quantitative** data on stakeholder familiarity with SaMD regulations and **qualitative** perspectives on perceived gaps or barriers (Challen et al., 2019). Interviews will follow a **semi-structured** format, exploring issues such as:

- Experience with **software or AI device approval**.
- Concerns about **bias, privacy, or algorithmic drift**.
- Views on **regulatory updates** or guidelines needed for AIaMD (Maidment, 2023).

7.4 Comparative Analysis

The **comparative analysis** will merge documentary findings and stakeholder insights to address:

- Areas of **alignment** (e.g., if both law and stakeholders identify cybersecurity as a key gap).
- Areas of **divergence** (e.g., if guidelines seem sufficient but stakeholders are unaware of them, indicating an implementation gap).

Priority gaps will be determined based on **frequency** (how often stakeholders mention them) and **normative significance** (importance in international frameworks) (WHO, 2017; FDA, 2021).

7.5 Recommendations and Operational Compliance Tool

Finally, the study will produce:

1. **Policy Recommendations**: Suggested **updates to existing regulations** or creation of new **SAHPRA guidelines**. For instance, adopting FDA-inspired *Predetermined Change Control Plans* (PCCP) could streamline **continuous learning updates** while maintaining safety (FDA, 2021).

2. **Operational Checklist**: A concise **compliance tool** for regulators and manufacturers, covering key items such as **bias testing, drift monitoring, cybersecurity protocols, performance validation,** and **post-market reporting**. This tool will be grounded in ISO 13485:2016 and IEC 62304:2006 standards, ensuring **international alignment** (ISO 13485:2016; IEC 62304:2006).

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8. Expected Outcomes

1. **Comprehensive Gap Analysis**: A **mapped comparison** of South African regulations against FDA, EU MDR, ISO, WHO guidelines, highlighting **specific AI-related shortcomings** (FDA, 2021; European Commission, 2017; WHO, 2017).

2. **Stakeholder Perceptions**: Insight into local viewpoints, revealing **practical barriers** (e.g., capacity constraints at SAHPRA, low AI awareness in IRBs) and **areas of consensus** (Challen et al., 2019).

3. **Prioritized Recommendations**: Actionable steps to **update regulatory texts**, improve post-market surveillance, and enhance transparency and safety of AIaMD.

4. **Compliance Tool**: A practical checklist or framework, enabling developers and regulators to **assess readiness** for AIaMD approvals and ensuring **alignment with best practices** (ISO 13485:2016).

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9. Significance of the Study

9.1 Regulatory Impact

The research provides **clear, evidence-based strategies** for improving South Africa's oversight of SaMD/AIaMD (Maidment, 2023). A modernized framework can **reduce patient risk**, support clinical adoption of AI tools, and **boost local healthtech innovation**.

9.2 Healthcare and Patient Safety

By emphasizing **algorithmic transparency**, **bias mitigation**, and **robust validation**, the proposed measures aim to **protect patients** while fostering trust in AI-driven healthcare (Challen et al., 2019).

9.3 Academic Contribution

As an **LLM thesis**, this study adds to the growing body of legal scholarship on **AI governance** in low- and middle-income country contexts. Its **hybrid approach** (doctrinal + empirical) may serve as a model for other jurisdictions.

9.4 Economic and Societal Benefits

A stable, **innovation-friendly regulatory environment** can spark **investment** in local AI healthtech solutions. Ensuring that these solutions meet **global standards** fosters both **domestic healthcare improvements** and **export opportunities** (Ravishankar et al., 2020).

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